

Heng Jin

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Research Portfolio: [heng_jin.github.io](https://github.com/hengjin)

EDUCATION

Southern University of Science and Technology (SUSTech)

Shenzhen, China

- M.Eng. in Computer Technology

Sept. 2024 - Expected Jun. 2027

Hefei University of Technology (HFUT)

Hefei, China

- B.Sc. in Applied Physics

Sept. 2019 - Jun 2023

RESEARCH EXPERIENCE

Graduate Student Research Assistant

2024 – Present

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

Shenzhen, China

Supervised by Prof. Xiangxin Li and Prof. Guanglin Li (Director, SIAIT, CAS)

- Developed an adaptive low-rank matrix completion framework (ALF-SVT) for robust HD-sEMG signal recovery under mixed-channel corruption and motion-interference conditions, improving signal reliability for upper-limb prosthetic interaction.
- Developed and evaluated a vision-guided shared-control framework for a full-arm prosthesis, integrating EMG-based user intention, RGB-D environmental perception, and autonomous motion assistance for coordinated reaching and grasping; validated the system with able-bodied participants and a transhumeral amputee.
- Conducted exploratory cortico-muscular coupling analysis in vision-guided prosthetic interaction tasks to investigate neural-muscular coordination patterns and human-factor-related changes in user control burden under different control modes.
- Developed a temporal attention-guided Mixture-of-Modality-Experts (MoME) fusion framework for robust EMG-FMG multimodal hand gesture recognition, achieving high-accuracy gesture classification on amputee subjects and superior noise robustness against conventional multimodal fusion strategies.

PUBLICATIONS (* indicates equal contribution)

Journal Publications

- **Jin, H.**, Tofunmi Oyemakinde, T., Liu, Y., Huang, J., Sun, J., Liu, Z., Li, G., Li, X., 2026. An Adaptive Signal Recovery Framework Based on Low-Rank Matrix Completion for High-Density sEMG. *IEEE Trans. Instrum. Meas.* 75, 1–12.
- Chen, X., Liu, Y., **Jin, H.**, Oyemakinde, T. T., Li, G., Li, X., 2026. AdaWGAN: Data Augmentation for Few-Shot HD-sEMG Gesture Recognition Using Single-Trial Data. *IEEE J. Biomed. Health Inform.*

Journal Manuscripts Under Review

- Liu, Y.*, **Jin, H.***, Zhu, Y., Chen, X., Zheng, Y., Tian, L., Deng, H., Li, G., & Li, X. 2026. Vision-guided full-arm prosthetic control framework based on EMG and RGB-D fusion. *IEEE Trans. Neural Syst. Rehabil. Eng.* (***Under review***).

Manuscripts in Preparation

- **Jin, H.***, Oyemakinde, T. T.*, et al. 2026. Human Factors and Neural-Muscular Coordination in Vision-Guided Prosthetic Interaction: A Cortico-Muscular Coupling Study.
- Oyemakinde, T. T.*, **Jin, H.***, et al. 2026. Temporal Attention-Guided Mixture-of-Modality-Experts Fusion for Robust EMG-FMG Gesture Recognition.

Conference Presentations

- **Jin, H.**, Oyemakinde, T.T., Liu, Y., Huang, J., Sun, J., Li, X., 2026. Adaptive Recovery Framework of High-density Surface EMG Signal Based on Low-rank Matrix Completion. In: 2026 Rehabilitation Engineering Conference (REC2026), China. Poster Presentation.

- **Jin, H.**, Li, X., 2026. Low-Rank Matrix Completion Based Adaptive Recovery Framework for High-Density sEMG Signals. In: 2026 Annual Conference of Chinese Society of Biomedical Engineering (BME2026), China. Poster Presentation.
- Oyemakinde, T.T., **Jin, H.**, Liu, Y., 2026. Explainable AI as a New Criterion for Feature Selection in Multimodal EMG-FMG Gesture Recognition. In: 2026 Rehabilitation Engineering Conference (REC2026), China. Poster Presentation.

Patents

- Li, X., Liu, Y., Zhu, Y., **Jin, H.**, Tian, L., Zheng, Y., Li, G., 2025. Remote Bionic Pulse Diagnosis System Based on Motion Intention Recognition. Chinese Invention Patent, Application No. PCT/CN2025/142485.

SKILLS & INTERESTS

Research: Biosignal Processing, Multimodal Human Sensing, Shared Robotic Control, Human–Robot Interaction

Programming: Python, MATLAB, PyTorch

Experimental: EMG/HD-sEMG, EEG, FMG, RGB-D based robotic interaction, human-subject experiments.

AWARDS

Second Prize, 2026 World Robot Contest – BCI Controlled Robot Contest	2026
First-Class Academic Scholarship, Southern University of Science and Technology	2025
Third Prize, National College Physics Experiment Competition, Hefei University of Technology	2022